



CS & IT ENGINEERING

Discrete Mathematics

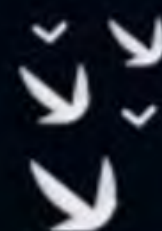
Mathematical logic

Lecture No.- 06

By- Satish Sir



Recap of Previous Lecture



Topic

Logic Part-05



Topics to be Covered



Topic

Logic Part-06



$$1. \exists x [p(x) \wedge q(x)] \rightarrow \exists x p(x) \wedge \exists x q(x) \quad \checkmark$$

$$2. \exists x [p(x) \vee q(x)] \leftrightarrow \exists x p(x) \vee \exists x q(x) \quad \checkmark$$

$$3. \forall x [p(x) \wedge q(x)] \leftrightarrow \forall x p(x) \wedge \forall x q(x) \quad \checkmark$$

$$4. \forall x [p(x) \vee q(x)] \leftarrow \forall x p(x) \vee \forall x q(x)$$

$$5. \forall x [p(x) \rightarrow q(x)] \rightarrow (\forall x p(x) \rightarrow \forall x q(x))$$

$$6. \forall x [p(x) \leftrightarrow q(x)] \Rightarrow (\forall x p(x) \leftrightarrow \forall x q(x))$$

2.

$$\exists x [P(x) \vee Q(x)] \iff$$

$$\exists x P(x) \vee \exists x Q(x)$$

I
one box
must be
True



✓



✓

✓



I

✓

I



✓

✓

I

✓



✓

✓

check:

$$\forall x [p(x) \vee q(x)] \not\Rightarrow \forall x p(x) \vee \forall x q(x)$$

True.

$$\left\{ \begin{array}{l} \boxed{T \vee F} \quad T \\ \quad \wedge \\ \boxed{F \vee T} \quad T \\ \quad \wedge \\ \boxed{T \vee F} \quad T \end{array} \right.$$

$$\begin{array}{c} \boxed{T} \\ \wedge \\ \boxed{F} \\ \wedge \\ \boxed{T} \end{array} \vee \begin{array}{c} \boxed{F} \\ \wedge \\ \boxed{T} \\ \wedge \\ \boxed{F} \end{array}$$

$$T \vee F$$

$$\forall x \rightarrow \text{AND}$$

$$\exists x \rightarrow \text{OR}$$

$$\forall x (p(x) \vee q(x)) \not\Rightarrow \forall x p(x) \vee \forall x q(x)$$

$$\forall x (p(x)) \vee \forall x q(x) \rightarrow \forall x (p(x) \vee q(x))$$

Check:

$$\forall x P(x) \vee \forall x Q(x) \rightarrow \forall x [P(x) \vee Q(x)] \text{ (valid)}$$



∨

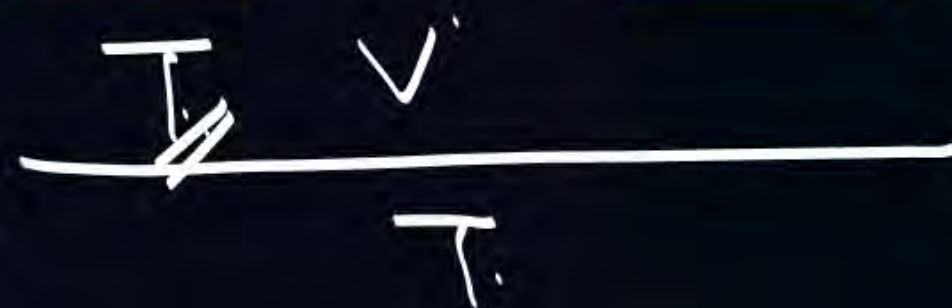


T

^



T



Cases:

T ∨ — ✓
 — ∨ T
 T ∨ T

$$\forall x [p(x) \rightarrow Q(x)] \rightarrow \forall x p(x) \rightarrow \forall x Q(x)$$

$$\begin{array}{l} \text{True} \left\{ \begin{array}{l} \boxed{\text{True} \rightarrow \text{True}} \text{ True} \\ \quad \wedge \\ \boxed{\text{False} \rightarrow \text{True}} \text{ True} \end{array} \right. \end{array}$$

$$\begin{array}{ccc} \boxed{\begin{array}{c} \text{True} \\ \wedge \\ \text{False} \end{array}} & \rightarrow & \boxed{\wedge} \\ \text{False} & \rightarrow & \end{array}$$

$$D: \{1, 2\} \quad \exists x [p(x) \wedge Q(x)] \rightarrow \exists x p(x) \wedge \exists x Q(x)$$

$$x=1: \boxed{p(1) \wedge Q(1)}$$

$$x=2: \boxed{p(2) \wedge Q(2)}$$

$$\boxed{\begin{matrix} p(1) \\ \vee \\ p(2) \end{matrix}}$$

$$\boxed{\begin{matrix} Q(1) \\ \vee \\ Q(2) \end{matrix}}$$

use Type-1
& check validity?

$$\underbrace{(p(1) \wedge Q(1)) \vee (p(2) \wedge Q(2))}_{\text{F}} \rightarrow \underbrace{(p(1) \vee p(2)) \wedge (Q(1) \vee Q(2))}_{\text{T}}$$

Case 1:

$$\begin{matrix} \downarrow & \downarrow \\ \text{T} & \text{T} \end{matrix}$$

T ∨ F

$$\begin{matrix} \downarrow \\ \text{T} \vee \text{F} \\ \hline \text{T} \end{matrix}$$

$$\begin{matrix} \downarrow \\ \text{T} \vee \text{F} \\ \hline \text{T} \end{matrix}$$



T

will be True @ $x = 3, 5$

D: 2.

$$p(x): x^2 - 8x + 15 = 0$$

$q(x): x$ is odd.

$$r(x): x > 0$$

$$a) \forall x [p(x) \rightarrow q(x)]$$

for all values of x , when it is True for $p(x)$ then it odd.
 $x = 3, 5$

$$\forall x [p(x) \rightarrow q(x)]$$

T.	{	$p(0) \rightarrow$		T.
		$\neg$	$\rightarrow$	
		$\wedge$		
		$p(1) \rightarrow$		T.
		$\neg$	$\rightarrow$	
T.	{	$\wedge$		
		$p(2) \rightarrow$		T.
		$\neg$	$\rightarrow$	
		$\wedge$		
		$p(3) \rightarrow Q(3)$		T.
		$\neg$	$\rightarrow$	
		$\wedge$		
		$p(5) \rightarrow Q(5)$		T.
		$\neg$	$\rightarrow$	
		$\wedge$		

will be True @ $x = 3, 5$

$$p(x): x^2 - 8x + 15 = 0$$

$q(x): x$ is odd.

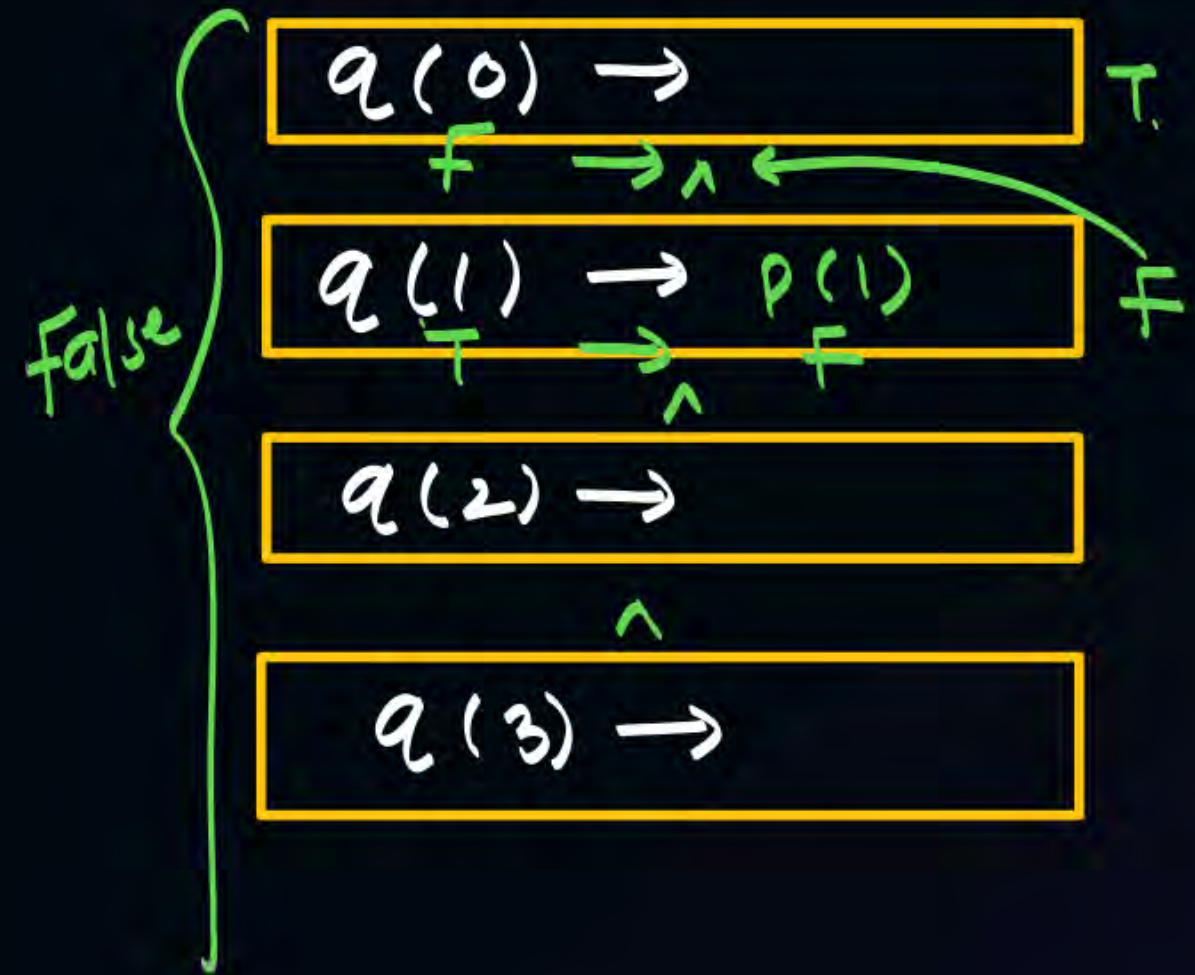
$$r(x): x > 0$$

$$b) \forall x [q(x) \rightarrow p(x)]$$

D: 2



$$\forall x (q(x) \rightarrow p(x))$$



will be True @ $x = 3, 5$

$$p(x): x^2 - 8x + 15 = 0$$

$q(x): x$ is odd.

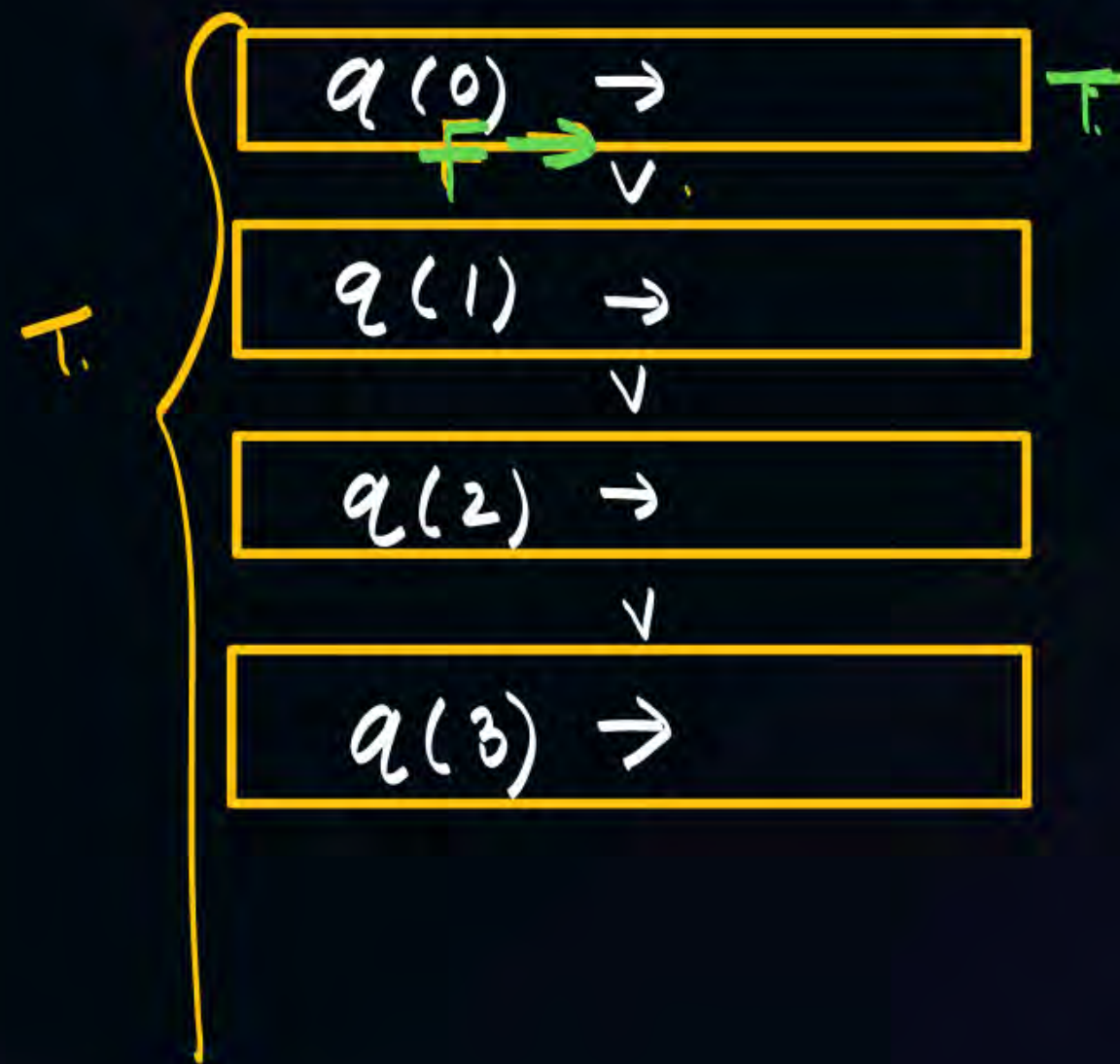
$$r(x): x > 0$$

$$3) \exists x [q(x) \rightarrow p(x)]$$

D: $\mathbb{Z}$



$$\exists x [q(x) \rightarrow p(x)]$$



will be True @ $x = 3, 5$

$$p(x): x^2 - 8x + 15 = 0$$

$q(x): x$ is odd.

$$r(x): x > 0$$

4) $\exists x [R(x) \rightarrow p(x)]$

5) $\forall x [\neg q(x) \rightarrow \neg p(x)]$

6) $\exists x [p(x) \rightarrow (q(x) \wedge R(x))]$

7) $\forall x [(p(x) \vee q(x)) \rightarrow R(x)]$



2 mins Summary



Topic

One

Logic Part-06

Topic

Two

Topic

Three

Topic

Four

Topic

Five



THANK - YOU